Topic and contents

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Topic 10 – Stochastic integration and SDEs

Random Walk

- Time interval $[0, T]$. Let $\Delta t = T/N$ and define the time steps:

$$t_n = n\Delta t \quad \text{for } n = 0, 1, 2, \ldots, N.$$

- A "walker" has position X_n at time t_n, and its next position is

$$X_{n+1} = X_n + \Delta X_n \quad \text{for } n = 0, 1, 2, \ldots, N-1.$$

- Revision: Random variables X, Y and constants a, b, c.
  - Expected value is linear

$$\mathbb{E}\left[aX + bY + c\right] = a\mathbb{E}\left[X\right] + b\mathbb{E}\left[Y\right] + c.$$

  - Variance

$$\mathbb{V}ar\left[X\right] := \mathbb{E}\left[(X - \mathbb{E}\left[X\right])^2\right] = \mathbb{E}\left[X^2\right] - (\mathbb{E}\left[X\right])^2.$$

  - If X and Y are independent, then
    - $\mathbb{E}\left[XY\right] = \mathbb{E}\left[X\right]\mathbb{E}\left[Y\right]$,
    - $\mathbb{V}ar\left[aX + bY\right] = a^2\mathbb{V}ar\left[X\right] + b^2\mathbb{V}ar\left[Y\right]$.

Random Walk model

- Model: Assume the increments ΔX_n satisfy

$$\Delta X_n = \mu\,\Delta t + \sigma\,(\Delta t)^\gamma Z_n,$$

where μ, σ and γ are constants, and $Z_0, Z_1, \ldots, Z_{N-1}$ are independent standard normal random variables. Hence,

$$\mathbb{E}\left[Z_n\right] = 0 \quad \text{and} \quad \mathbb{E}\left[Z_n Z_j\right] = \delta_{nj} := \begin{cases} 1, & j = n, \\ 0, & j \neq n. \end{cases}$$

- $\mathbb{E}\left[\Delta X_n\right] = \mu\,\Delta t$ (drift)
$\mathbb{E}\left[\Delta X_n \Delta X_j\right] = \mathbb{E}\left[(\mu\Delta t + \sigma(\Delta t)^\gamma Z_n)(\mu\Delta t + \sigma(\Delta t)^\gamma Z_j)\right] =$
$\mathbb{E}\left[\mu^2\Delta t^2 + \mu\sigma(\Delta t)^{\gamma+1}(Z_j + Z_n) + \sigma^2(\Delta t)^{2\gamma} Z_n Z_j\right] = \mu^2\,\Delta t^2 + \sigma^2(\Delta t)^{2\gamma}\delta_{nj}$.
- The final position X_N satisfies $X_N - X_0 = \sum_{n=0}^{N-1} \Delta X_n$. Therefore,
  - $\mathbb{E}\left[X_N - X_0\right] = N\mu\,\Delta t = \mu T$,
  - $\mathbb{V}ar\left[X_N - X_0\right] = \sum_{n=0}^{N-1} \mathbb{V}ar\left[\Delta X_n\right] = N\sigma^2(\Delta t)^{2\gamma} = \sigma^2 T(\Delta t)^{2\gamma-1}$.

Limit behavior and a stock model

- If the initial position $X_0 = x_0$ is known with certainty, then
  - $\mathbb{E}[X_N] = x_0 + \mu T$,
  - $$\mathbb{V}ar\,[X_N - x_0] \to \begin{cases} 0, & \gamma > 1/2, \\ \sigma^2 T, & \gamma = 1/2, \\ \infty, & \gamma < 1/2, \end{cases} \quad \text{as} \quad \Delta t \to 0.$$
- A natural choice is $\gamma = 1/2$, which gives a finite, non-zero limiting variance. Hence,

$$X_{n+1} = X_n + \Delta X_n = X_n + \mu\Delta t + \mu\Delta t + \sigma\sqrt{\Delta t}\, Z_n.$$

- For a stock price S_n, the size of the change ΔS_n should depend on the current stock price S_n. We therefore model the relative change:

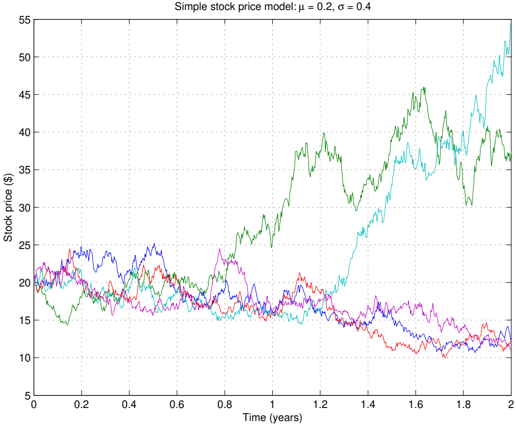
$$\frac{\Delta S_n}{S_n} = \mu\,\Delta t + \sigma\sqrt{\Delta t}\, Z_n.$$

Using $\Delta S_n = S_{n+1} - S_n$, then $S_{n+1} = S_n\left(1 + \mu\Delta t + \sigma\sqrt{\Delta t}\, Z_n\right)$.

Iterating the recursion gives

$$S_{n+1} = S_0 \prod_{j=0}^{n} \left(1 + \mu \Delta t + \sigma \sqrt{\Delta t}\, Z_j\right), \qquad n \geq 0.$$

`Ex.: S0 = 20, Dt = 1/365, Zn=randn(N,5), generating 5 samples`



MATLAB `stock1.m`; Python `stock1.py`

Wiener process

A Wiener process, or standard Brownian motion, is the random process that models the uncertainty in geometric Brownian motion. It is denoted by $W(t)$ on the interval $[0, T]$ (or $(W_t)_{0 \le t \le T}$) and satisfies:

1. The sample paths $t \mapsto W(t)$ are continuous almost surely.
2. $W(0) = 0$.
3. $W(t) - W(s) \sim \mathcal{N}(0, t-s)$ for $0 \le s < t \le T$.
4. The increments $W(t) - W(s)$ and $W(v) - W(u)$ are independent over disjoint time intervals.

Hence, over a small time interval of length Δt,

$$W(t + \Delta t) - W(t) \approx \sqrt{\Delta t}\, Z, \quad Z \sim \mathcal{N}(0, 1).$$

On the time grid $t_j = j\Delta t$, $j = 0, 1, \ldots, N$, $\Delta t = T/N$, a Wiener path can therefore be simulated recursively by

$$W_0 = 0, \qquad W_{j+1} = W_j + \sqrt{\Delta t}\, Z_j,$$

where $Z_0, Z_1, \ldots, Z_{N-1}$ are independent standard normal random variables.

Ito integral

Let $\Delta t = T/N$ and $t_n = n\Delta t$ for $n \geq 0$. The Itô integral

$$\int_0^T f\big(W(t), t\big)\, dW(t) = \lim_{\Delta t \to 0} \sum_{n=0}^{N-1} f\big(W(t_n), t_n\big)\big(W(t_{n+1}) - W(t_n)\big).$$

Since $f(W(t_n), t_n)$ depends only on the Brownian path up to time t_n, it is independent of the future increment $W(t_{n+1}) - W(t_n)$. Therefore,

$$\begin{aligned} &\mathbb{E}\left[f\big(W(t_n), t_n\big)\big(W(t_{n+1}) - W(t_n)\big)\right] \\ &\qquad = \mathbb{E}\left[f\big(W(t_n), t_n\big)\right] \underbrace{\mathbb{E}\left[W(t_{n+1}) - W(t_n)\right]}_{=0}. \end{aligned}$$

The probability distribution of an Itô integral can be investigated numerically using discrete-time simulations of the Wiener process.

Ito integral - Example

MATLAB `wiener1.m, wiener2.m`; Python `wiener1.py, wiener2.py`

Show that

$$\int_0^T W(t)\, dW(t) = \tfrac{1}{2} W^2(T) - \tfrac{1}{2} T.$$

Solution.

$$\begin{aligned}
\int_0^T W(t)\, dW(t) &= \lim_{\Delta t \to 0} \sum_{n=0}^{N-1} W(t_n) \left(W(t_{n+1}) - W(t_n) \right) \\
&= \frac{1}{2} \lim_{\Delta t \to 0} \sum_{n=0}^{N-1} \left(W^2(t_{n+1}) - W^2(t_n) - (W(t_{n+1}) - W(t_n))^2 \right) \\
&= \frac{1}{2} \lim_{\Delta t \to 0} \left(W^2(t_N) - W^2(t_0) - \sum_{n=0}^{N-1} (W(t_{n+1}) - W(t_n))^2 \right) \\
&= \frac{1}{2} W^2(T) - \frac{1}{2} \lim_{\Delta t \to 0} \left(\sum_{n=0}^{N-1} (W(t_{n+1}) - W(t_n))^2 \right)
\end{aligned}$$

Let $\Delta W_n = W(t_{n+1}) - W(t_n)$. Since $\Delta W_n \sim \mathcal{N}(0, \Delta t)$,

$$\mathbb{E}\left[\sum_{n=0}^{N-1} (\Delta W_n)^2\right] = \sum_{n=0}^{N-1} \mathbb{E}\left[(\Delta W_n)^2\right] = \sum_{n=0}^{N-1} \mathbb{V}ar\left[\Delta W_n\right] = \sum_{n=0}^{N-1} \Delta t = N\Delta t = T.$$

Moreover, $\frac{1}{\sqrt{\Delta t}}\Delta W_n \sim \mathcal{N}(0,1)$ for $n \geq 0$, and therefore $\mathbb{V}ar\left[\left(\frac{1}{\sqrt{\Delta t}}\Delta W_n\right)^2\right] = 2$ (chi-squared distribution).

$$\mathbb{V}ar\left[\sum_{n=0}^{N-1} (\Delta W_n)^2\right] = \sum_{n=0}^{N-1} \mathbb{V}ar\left[(\Delta W_n)^2\right] = (\Delta t)^2 \sum_{n=0}^{N-1} \mathbb{V}ar\left[\left(\frac{1}{\sqrt{\Delta t}}\Delta W_n\right)^2\right]$$

$$= (\Delta t)^2 \sum_{n=0}^{N-1} 2 = 2N(\Delta t)^2 = 2T\Delta t \to 0 \quad \text{as } \Delta t \to 0.$$

Hence,

$$\lim_{\Delta t \to 0} \left(\sum_{n=0}^{N-1} (W(t_{n+1}) - W(t_n))^2\right) = T.$$

Stochastic differential equations (SDEs)

Suppose that the stochastic process $X(t)$ satisfies the SDE:

$$dX(t) = f(X(t), t)\, dt + g(X(t), t)\, dW(t), \qquad X(0) = X_0, \qquad (1)$$

for $0 \leq t \leq T$.

Lemma (Itô's lemma)

Let $Y(t) = H(X(t), t)$, where $X(t)$ satisfies (1), and $H(x, t)$ is continuously differentiable in t and twice continuously differentiable in x. Then, with $H_t = \frac{\partial H}{\partial t}$, $H_x = \frac{\partial H}{\partial x}$ and $H_{xx} = \frac{\partial^2 H}{\partial x^2}$, then

$$dY(t) = \left(H_t(X(t), t) + \frac{1}{2}[g(X(t), t)]^2 H_{xx}(X(t), t) \right) dt + H_x(X(t), t)\, dX(t).$$

Compare with standard calculus: $dY = H_t\, dt + H_x\, dx$.

Geometric Brownian motion (GBM)

In the Black–Scholes model, the asset price $S(t)$ is assumed to follow a GBM (the logarithm of the process follows an ordinary Brownian motion):

$$\frac{dS(t)}{S(t)} = \mu\, dt + \sigma\, dW(t) \qquad \Longleftrightarrow \qquad dS(t) = \mu S(t)\, dt + \sigma S(t)\, dW(t),$$

where μ is the drift, $\sigma > 0$ is the volatility, $W(t)$ is a standard Wiener process, and $S(0) = S_0 > 0$.

Let $Y(t) = H(S(t)) := \log(S(t)/S_0)$. Since $H'(S) = 1/S$ and $H''(S) = -1/S^2$, Itô's lemma gives

$$dY(t) = \frac{1}{2} H''(S(t))\big(\sigma S(t)\big)^2\, dt + H'(S(t))\, dS(t) = \big(\mu - \sigma^2/2\big)\, dt + \sigma\, dW(t).$$

Integrating from 0 to t,

$$\log(S(t)/S_0) = \big(\mu - \sigma^2/2\big)\, t + \sigma W(t).$$

Therefore,

$$S(t) = S_0 \exp\big[\big(\mu - \sigma^2/2\big)\, t + \sigma W(t)\big]\,.$$

Lognormal distribution of $S(t)$

Since $W(t) \sim \mathcal{N}(0, t)$, we have

$$\log\left(S(t)/S_0\right) \sim \mathcal{N}\left(\left(\mu - \tfrac{1}{2}\sigma^2\right) t,\ \sigma^2 t\right).$$

Hence, for $0 < a < b$,

$$\begin{aligned}
\text{Prob}\left(a < S(t) < b\right) &= \text{Prob}\left(\log(a/S_0) < \log(S(t)/S_0) < \log(b/S_0)\right) \\
&= \frac{1}{\sqrt{2\pi t}\,\sigma} \int_{\log(a/S_0)}^{\log(b/S_0)} \exp\left(-\frac{\left(x - (\mu - \frac{1}{2}\sigma^2)t\right)^2}{2\sigma^2 t}\right) dx \\
&= \frac{1}{\sqrt{2\pi t}\,\sigma} \int_a^b \frac{1}{y} \exp\left(-\frac{\left(\log(y/S_0) - (\mu - \frac{1}{2}\sigma^2)t\right)^2}{2\sigma^2 t}\right) dy,
\end{aligned}$$

and therefore, $S(t)$ has a lognormal distribution, with PDF

$$\frac{1}{y\sigma\sqrt{2\pi t}} \exp\left(-\frac{\left(\log(y/S_0) - \left(\mu - \frac{1}{2}\sigma^2\right) t\right)^2}{2\sigma^2 t}\right), \qquad y > 0.$$

Black–Scholes model: construction of a riskless portfolio

Consider a self-financing portfolio (you don't inject or withdraw cash) consisting of one option (value $V(S,t)$) and $f(S_t,t)$ short positions in the underlying asset ($\Pi(S,t) = V(S,t) - f(S,t)\,S$).
Over an infinitesimal time interval, its change in value is

$$d\Pi = dV - f(S,t)\,dS.$$

By Itó's Lemma, as S follows geometric Brownian motion ($dS = \mu S\,dt + \sigma S\,dW$),

$$dV = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + V_S\,dS,$$

where $V_t = \partial V/\partial t$, $V_S = \partial V/\partial S$, and $V_{SS} = \partial^2 V/\partial S^2$. So

$$d\Pi = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)dt + \left(V_S - f\right)dS.$$

To eliminate all risk (i.e., stochasticity), choose: $f(S,t) = V_S(S,t)$. Then the portfolio evolves deterministically:

$$d\Pi = \left(V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS}\right)\,dt.$$

Black-Scholes PDE

By the no-arbitrage principle, a locally riskless portfolio must earn the risk-free rate r. Hence,

$$d\Pi = r\Pi\, dt \Longrightarrow (V_t + \tfrac{1}{2}\sigma^2 S^2 V_{SS})\, dt = r(V - SV_S)\, dt,$$

which gives the Black–Scholes PDE:

$$V_t + rSV_S + \tfrac{1}{2}\sigma^2 S^2 V_{SS} = rV.$$

Key observations: The PDE involves the risk-free rate r and the volatility σ, but not on the stock drift μ of the stock. This reflects a fundamental principle: *option pricing does not depend on the expected return of the underlying asset*.

Model assumptions:

- The asset price follows geometric Brownian motion: $\dfrac{dS}{S} = \mu\, dt + \sigma\, dW$
- r and σ are known constants.
- There are no transaction costs or taxes, there are no arbitrage opportunities.
- The underlying asset pays no dividends.
- Continuous trading and short selling are allowed.

Summary: The Black–Scholes PDE

The value $V(S,t)$ of a European option satisfies the Black–Scholes PDE:

$$\frac{\partial V}{\partial t} + rS\,\frac{\partial V}{\partial S} + \tfrac{1}{2}\sigma^2 S^2\,\frac{\partial^2 V}{\partial S^2} = rV \qquad \text{for } 0 < S < \infty, \quad 0 < t < T.$$

Recall that, a European call (put) option gives its holder the right, but not the obligation, to buy (sell) the underlying asset at the strike price K at the expiry time T. Also, recall that

$$\begin{aligned} S &= \text{price of underlying asset,} \\ t &= \text{time,} \\ r &= \text{risk-free interest rate,} \\ \sigma &= \text{volatility of underlying asset,} \\ c(S,t) &= \text{value of the European call option at time } t, \\ p(S,t) &= \text{value of the European put option at time } t. \end{aligned}$$

Terminal and boundary conditions

To determine a unique solution of the Black–Scholes PDE, we must specify a terminal condition at $t = T$ and suitable boundary conditions in S.

European call option: $V(S,t) = c(S,t)$

$$\begin{aligned} c(S,T) &= (S-K)^+ = \max(S-K,0), && \text{terminal payoff,} \\ c(0,t) &= 0, && 0 \le t \le T, \\ c(S,t) &\sim S - Ke^{-r(T-t)}, && \text{as } S \to \infty. \end{aligned}$$

European put option: $V(S,t) = p(S,t)$

$$\begin{aligned} p(S,T) &= (K-S)^+ = \max(K-S,0), && \text{terminal payoff,} \\ p(0,t) &= Ke^{-r(T-t)}, && 0 \le t \le T, \\ p(S,t) &\to 0, && \text{as } S \to \infty. \end{aligned}$$

Interpretation.

- If $S = 0$, the call is worthless.
- If $S = 0$, the put pays K at expiry and is therefore worth $Ke^{-r(T-t)}$ at time t.

Change of variables to simplify the model

- New independent variables $\Longrightarrow$ simplified Black–Scholes PDE

$$y = \log(S/K) = \text{log moneyness} \quad \Longleftrightarrow \quad S = Ke^y,$$
$$\tau = T - t = \text{time to expiry} \quad \Longleftrightarrow \quad t = T - \tau.$$

- PDE for $V(S,t)$: $\frac{\partial V}{\partial t} + rS\frac{\partial V}{\partial S} + \frac{1}{2}\sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} = rV$
- Function $V(S,t) = U(y,\tau)$
- Partial derivatives
  - $\dfrac{\partial V}{\partial t} = \dfrac{\partial U}{\partial \tau}\dfrac{\partial \tau}{\partial t} = -\dfrac{\partial U}{\partial \tau}$
  - $\dfrac{\partial V}{\partial S} = \dfrac{\partial U}{\partial y}\dfrac{\partial y}{\partial S} = \dfrac{1}{S}\dfrac{\partial U}{\partial y}$
  - $\dfrac{\partial^2 V}{\partial S^2} = \dfrac{\partial}{\partial S}\left(\dfrac{1}{S}\dfrac{\partial U}{\partial y}\right) = -\dfrac{1}{S^2}\dfrac{\partial U}{\partial y} + \dfrac{1}{S}\dfrac{\partial}{\partial y}\left(\dfrac{\partial U}{\partial y}\right)\dfrac{\partial y}{\partial S} = \dfrac{1}{S^2}\left(\dfrac{\partial^2 U}{\partial y^2} - \dfrac{\partial U}{\partial y}\right)$
- PDE for $U(y,\tau)$: $-\frac{\partial U}{\partial \tau} + \left(r - \frac{1}{2}\sigma^2\right)\frac{\partial U}{\partial y} + \frac{1}{2}\sigma^2\frac{\partial^2 U}{\partial y^2} = rU$

Change of variables to simplify the model (continued)

- Domain
  - $0 < S < \infty \Longrightarrow -\infty < y = \log(S/K) < \infty$. At $S = K$, $y = 0$.
  - Time: $t \in [0, T] \iff \tau \in [0, T]$. At (expiry) $t = T$, $\tau = 0$.
- For a call option $V(S, t) = c(S, t)$ so $U(y, \tau) = c(Ke^y, T - \tau)$,
  - IC: For $-\infty < y < \infty$,

$$U(y, 0) = \max\{Ke^y - K, 0\} = K \max\{e^y - 1, 0\}.$$

  - BCs: For $0 < \tau < T$,

$$U(y, \tau) \to 0 \text{ as } y \to -\infty, \text{ and } U(y, \tau) \approx Ke^y - Ke^{-r\tau} \text{ as } y \to \infty.$$

- For a put option $V(S, t) = p(S, t)$ so $U(y, \tau) = p(Ke^y, T - \tau)$,
  - IC: For $-\infty < y < \infty$,

$$U(y, 0) = \max\{K - Ke^y, 0\} = K \max\{1 - e^y, 0\}.$$

  - BCs: For $0 < \tau < T$,

$$U(y, \tau) = Ke^{-r\tau} \text{ as } y \to -\infty, \text{ and } U(y, \tau) \to 0 \text{ as } y \to \infty.$$

Put-Call parity relation

Consider two portfolios involving European options with the same strike price K and expiry time T:

1. Portfolio Π_1 consists of one share of the underlying asset and one European put option: $\Pi_1(S,t) = S + p(S,t)$.
2. Portfolio Π_2 consists of one European call option and a risk-free bond (cash) worth the present value of the strike price:

$$\Pi_2(S,t) = c(S,t) + Ke^{-r(T-t)}.$$

At expiry, $t = T$, the portfolio values are

$$\Pi_1(S,T) = S + \max(K - S, 0) = \max(K, S),$$
$$\Pi_2(S,T) = \max(S - K, 0) + K = \max(S, K).$$

Thus, both portfolios have the same payoff at maturity regardless of the asset price S, that is, $\Pi_1(S,T) = \Pi_2(S,T) \quad \text{for } 0 < S < \infty$.
By the no-arbitrage principle, their current values must therefore be equal:

$$\Pi_1(S,t) = \Pi_2(S,t) \quad \text{for } 0 < t < T.$$

This leads to the put-call parity relation: $S + p(S,t) = c(S,t) + Ke^{-r(T-t)}$.